

CSA1713 – Artificial Intelligence

Name:- B. Deepthi reddy

Reg No:- 192312288

Slot – C

4. VACUUM CLEANER PROBLEM

Code:

```
print("B. Deepthi Reddy")
print(192312288)
print("Vacuum Cleaner Problem")
A = input("Enter Room A status (Dirty/Clean): ")
B = input("Enter Room B status (Dirty/Clean): ")
print("\nInitial State")
print("Room A =", A)
print("Room B =", B)
print("\nVacuum starts at Room A")
if A.lower() == "dirty":
    print("Room A is Dirty")
    print("Cleaning Room A...")
    A = "Clean"
else:
    print("Room A is already Clean")
print("\nMove to Room B")
```

```

if B.lower() == "dirty":
    print("Room B is Dirty")
    print("Cleaning Room B...")
    B = "Clean"
else:
    print("Room B is already Clean")

print("\nFinal State")

print("Room A =", A)
print("Room B =", B)
print("\nAll rooms are clean.")

```

Output:

main.py	Run	Output
<pre> 1 print("B. Deepthi Reddy") 2 print(192312288) 3 print("Vacuum Cleaner Problem") 4 5 A = input("Enter Room A status (Dirty/Clean): ") 6 B = input("Enter Room B status (Dirty/Clean): ") 7 8 print("\nInitial State") 9 print("Room A =", A) 10 print("Room B =", B) 11 12 print("\nVacuum starts at Room A") 13 14 if A.lower() == "dirty": 15 print("Room A is Dirty") 16 print("Cleaning Room A...") 17 A = "Clean" 18 else: 19 print("Room A is already Clean") </pre>	Run	<pre> B. Deepthi Reddy 192312288 Vacuum Cleaner Problem Enter Room A status (Dirty/Clean): Clean Enter Room B status (Dirty/Clean): Dirty Initial State Room A = Clean Room B = Dirty Vacuum starts at Room A Room A is already Clean Move to Room B Room B is Dirty Cleaning Room B... Final State Room A = Clean </pre>

